

DILI — Drug AIH, Amiodarone, Statins & Checkpoint Hepatitis



1. Room 1: interface hepatitis histology

WHAT YOU SEE

Pinned slides: INTERFACE HEPATITIS, PLASMA CELL STAINING, ANA 1:320

WHAT IT ENCODES

Drug-induced autoimmune hepatitis shows the same histology as idiopathic AIH — interface hepatitis (lymphoplasmacytic infiltrate crossing the limiting plate), abundant plasma cells, and positive ANA/ASMA with elevated IgG. It is hepatocellular (ALT-dominant) in pattern.

BOARD TRAP

WRONG to call interface hepatitis + plasma cells + positive ANA automatically 'idiopathic AIH' — when a culprit drug is present (minocycline, nitrofurantoin, hydralazine, methyldopa, statins, anti-TNF), it is drug-induced AIH, which behaves differently on drug withdrawal.

2. Minocycline/hydralazine/methyldopa shadows

WHAT YOU SEE

Three dark figures: MINOCYCLINE, HYDRALAZINE, methyldopa

WHAT IT ENCODES

Classic drug-induced AIH offenders: minocycline (young patients on acne therapy, often with drug-induced lupus), hydralazine, methyldopa, nitrofurantoin, statins, and anti-TNF agents (infliximab/etanercept). Onset is typically after months of therapy.

BOARD TRAP

WRONG to overlook minocycline in a young acne patient with new hepatitis and arthralgias/positive ANA — this is the classic drug-induced AIH/lupus scenario, and stopping the drug (not lifelong immunosuppression) is the answer.

3. Rule out drug first

WHAT YOU SEE

PRIMARY AIH crossed out: RULE OUT DRUG FIRST

WHAT IT ENCODES

In a patient on a culprit drug presenting with AIH features, suspect DRUG-induced AIH first. Unlike idiopathic AIH, it resolves after stopping the drug ($\pm$ a short steroid course) and characteristically does NOT relapse after steroids are tapered off.

BOARD TRAP

WRONG to diagnose primary AIH and commit to lifelong immunosuppression without first stopping the drug — durable remission off steroids after dechallenge is the discriminator that proves it was drug-induced, not idiopathic.

4. Room 2: drug stopped, off the hook

WHAT YOU SEE

DRUG STOPPED 2 WEEKS AGO, OFF THE HOOK figure

WHAT IT ENCODES

Improvement after dechallenge (stopping the drug) is the strongest practical evidence for DILI causality and the key therapeutic maneuver. Most DILI improves within days to weeks of withdrawal; lack of improvement should prompt re-evaluation for another cause.

BOARD TRAP

WRONG to assume rapid resolution in every case — cholestatic DILI (e.g., amox-clav, vanishing bile duct syndrome) and amiodarone can take months to improve, so slow recovery after dechallenge does not refute the diagnosis.

5. Cholestatic pattern / amox-clav / pruritus

WHAT YOU SEE

CHOLESTATIC PATTERN, ALP/GGT up, PRURITUS, liver dripping bile

WHAT IT ENCODES

Cholestatic DILI shows elevated ALP and GGT ($R < 2$), pruritus, and jaundice. Amoxicillin-clavulanate is the prototypical cause (#1 idiosyncratic DILI) and characteristically presents days-to-weeks AFTER the antibiotic course is completed; anabolic steroids and estrogens cause bland cholestasis.

BOARD TRAP

WRONG to dismiss amox-clav because the course is over, or to call an isolated ALP rise 'cholestatic' without confirming a high GGT — a normal GGT points to a bony source, not cholestatic liver injury.

6. Room 3: months after stopping / amiodarone

WHAT YOU SEE

MONTHS AFTER STOPPING flame, AMIODARONE lamellar bodies

WHAT IT ENCODES

Amiodarone is highly lipophilic with a half-life of weeks-to-months and accumulates in tissue, so hepatotoxicity (like its pulmonary and thyroid toxicity) can appear or progress MONTHS after discontinuation. Injury can advance to cirrhosis despite stopping the drug.

BOARD TRAP

WRONG to exclude amiodarone because it was stopped months ago — unlike most drugs that improve on dechallenge, amiodarone persists in tissue and injury can continue or worsen long after withdrawal.

7. Amiodarone vs alcohol histology

WHAT YOU SEE

Screen: AMIODARONE lamellar bodies vs TRUE ALCOHOLIC lamellar bodies = NONE

WHAT IT ENCODES

Amiodarone causes phospholipidosis with characteristic lamellar (myeloid) inclusion bodies on electron microscopy, plus Mallory bodies and steatohepatitis that mimics alcoholic hepatitis on light microscopy. The lamellar inclusions are NOT seen in true alcoholic liver disease.

BOARD TRAP

WRONG to diagnose alcoholic hepatitis in a non-drinker on amiodarone with Mallory bodies and steatohepatitis — the lamellar inclusion bodies (and the drug history) are the discriminators; histology alone overlaps.

8. Room 4: statin 2.8x ULN / continue

WHAT YOU SEE

Runner: 2.8x ULN, SAFE in HCV/steatosis/CLD, CONTINUE/MONITOR

WHAT IT ENCODES

Statins cause a mild ALT rise that, when <3x ULN and asymptomatic, is managed by CONTINUING and monitoring (often adaptation). Statins are safe in chronic HCV, NAFLD/steatosis, and compensated chronic liver disease, and routine LFT monitoring is no longer required.

BOARD TRAP

WRONG to stop a statin for ALT 2.8x ULN (i.e., <3x ULN) or because the patient has stable chronic liver disease — this is the most-tested statin trap; you continue, since CV benefit outweighs the rare serious hepatotoxicity.

9. Patient stops statin (crossed)

WHAT YOU SEE

PATIENT STOPS STATIN with red X

WHAT IT ENCODES

Do NOT reflexively discontinue statins for a minor transaminase rise. Serious statin hepatotoxicity is rare (~1 in 100,000), and the cardiovascular benefit far outweighs it; stop only for ALT >3x ULN with symptoms/jaundice or signs of true hepatotoxicity.

BOARD TRAP

WRONG answer: discontinuing the statin (or denying it) over a trivial LFT bump — undertreating ASCVD risk to chase a benign, often adaptive ALT elevation is the error the question is testing.

10. Room 5: ANA 1:320 drug AIH

WHAT YOU SEE

ANA 1:320, NORMAL LOW vs HIGH titer, PRIMARY AIH crossed

WHAT IT ENCODES

A positive ANA (e.g., 1:320) is NOT specific for primary AIH — low-titer ANA occurs in healthy people, and in the context of a culprit drug it reflects drug-induced AIH. Correlate titer with clinical picture, IgG, ASMA, and histology rather than treating ANA alone.

BOARD TRAP

WRONG to equate a positive ANA with idiopathic AIH and start lifelong immunosuppression — in the right drug context it is drug-induced AIH that resolves on dechallenge; ANA positivity alone does not establish primary AIH.

11. Nivolumab / PD-1 checkpoint

WHAT YOU SEE

NIVOLUMAB / PD-1 vial, knight in armor

WHAT IT ENCODES

Immune checkpoint inhibitors — anti-PD-1 (nivolumab, pembrolizumab), anti-PD-L1 (atezolizumab), and anti-CTLA-4 (ipilimumab) — cause immune-mediated hepatitis as an immune-related adverse event (irAE). Risk is higher with CTLA-4 blockade and with combination therapy.

BOARD TRAP

WRONG to attribute new transaminitis in a cancer patient on immunotherapy to metastases or progression by default — checkpoint-inhibitor hepatitis is an irAE that must be graded and is treated by holding the drug and giving steroids, not just imaging.

12. Glucocorticoids effective

WHAT YOU SEE

GLUCOCORTICOIDS EFFECTIVE banner

WHAT IT ENCODES

For checkpoint-inhibitor hepatitis: hold the drug, and for high-grade (grade ≥3) injury give corticosteroids (e.g., prednisone/methylprednisolone 1–2 mg/kg/day), adding mycophenolate mofetil if steroid-refractory. This is one of the few DILI types with a specific immunosuppressive treatment.

BOARD TRAP

WRONG to manage severe checkpoint-inhibitor hepatitis with supportive care alone — unlike most idiosyncratic DILI (no specific antidote), high-grade irAE hepatitis specifically requires corticosteroids; withholding steroids is the error.

13. CD8+ T-knight warriors

WHAT YOU SEE

CD8+ T-KNIGHT WARRIORS attacking cells

WHAT IT ENCODES

Checkpoint inhibitors release the brakes on T cells, unleashing CD8+ cytotoxic T cells against hepatocytes and producing an autoimmune-like lobular/panlobular hepatitis. The mechanism explains why immunosuppression (steroids) is effective.

BOARD TRAP

WRONG to expect the classic autoantibody profile of AIH (ANA/ASMA) in checkpoint hepatitis — it is often autoantibody-negative T-cell-mediated injury, so a negative ANA does not exclude immune-mediated hepatitis in an immunotherapy patient.

14. Detective magnifier

WHAT YOU SEE

Detective with magnifying glass over the floor

WHAT IT ENCODES

DILI workup is detective work: establish the temporal relationship to drug start, document the dechallenge response, exclude alternative causes (viral/autoimmune/biliary), and identify the R-value pattern. RUCAM scoring structures this causality assessment.

BOARD TRAP

WRONG answer: deliberate rechallenge to 'confirm' causality — although positive rechallenge is the strongest evidence, it is contraindicated/potentially fatal; rely on temporality and dechallenge instead.